

Pointwise Bdd.

Uniform boundedness

Uniform boundedness principle.

X - Banach space

$\mathcal{A}$ = lin operators $\subseteq \underline{B(X, Y)}$

If $\mathcal{A}$ is pointwise bounded $\Rightarrow \mathcal{A}$ is uniformly ~~no~~ bdd.

Proof:

$\mathcal{A}$ is pointwise bounded.

Then for each $x \in X \quad \exists M_x > 0$

s.t $\| \underline{A} x \| \leq \underline{M}_x. \quad \underline{\underline{\forall A \in \mathcal{A}}}$.

$$E_n = \left\{ x \in X \mid \|Ax\| \leq n \quad \forall A \in \mathcal{A} \right\}$$

Note

$$X = \bigcup_{n=1}^{\infty} E_n$$

$$Ax = y$$

$$E_n = \left\{ A^{-1}y \mid \|y\| \leq n, \forall A \in \mathcal{A} \right\}$$

$$= \bigcap_{A \in \mathcal{A}} \underbrace{A^{-1}(B(0, n))}_{\text{closed}}$$

$\therefore E_n$'s are closed.

$$X = \bigcup_{n=1}^{\infty} E_n$$

By the Baire Category theorem

$$\exists k \text{ s.t. } \overline{E_k}^{\circ} \neq \emptyset$$

$$\Rightarrow E_k^{\circ} \neq \emptyset$$

Let $u \in E_k$ be an interior pt.

$$\therefore \exists \underline{r} > 0 \text{ s.t. } B(u, r) \subseteq E_k.$$

↓ open ball.

$$\text{Let } x \in X, \quad x \neq 0.$$

$$y = u + \frac{rx}{2\|x\|}$$

$$\|y - u\| = \frac{r}{2} < r$$

$$\Rightarrow y \in B(u, r) \subseteq E_k.$$

$$\therefore \|Ay\| \leq k$$

$$\therefore \boxed{\|A(u + \frac{rx}{2\|x\|})\| \leq k \quad \forall A \in \mathcal{A}.$$

Look at

$$\left\| A \left(\frac{rx}{2\|x\|} \right) \right\| \quad \forall A \in \mathcal{A}$$

$$= \left\| A \left(u + \frac{rx}{2\|x\|} - u \right) \right\|$$

$$\leq \left\| A \left(u + \frac{rx}{2\|x\|} \right) \right\| + \|Au\|$$

$$\leq \underline{k + k} \quad \forall A \in \mathcal{A}$$

$$\therefore \|Ax\| \leq \frac{4k}{r} \|x\|$$

$$\forall A \in \mathcal{A}$$

ie $\{ \|A\| : A \in \mathcal{A} \}$
is bdd subset of $\mathbb{R}$.

Consequences.

Let X be a B.S.

Y be a NLS.

$(A_n)_{n \in \mathbb{N}} \in B(X, Y)$

Suppose for each $x \in X$.

$(A_n \alpha)_{n \in \mathbb{N}}$ converges
to $A : X \rightarrow Y$

$$A \alpha = \lim_{n \rightarrow \infty} A_n \alpha.$$

Then A is a bdd
linear op.

(Banach Steinhaus Theo)

Theo. (Resonance Theo).

Let X, Y be NLS.

$E \subseteq X$. E is bdd
in $X \iff f(E)$ is bdd
in $K \quad \forall y \in X'$.

Proof: $E \subseteq X$ is bdd.

$\exists M > 0$ s.t.
 $\|x\| \leq M \quad \forall x \in E$

$$\begin{aligned} | \underline{f(x)} | &\leq \|f\| \|x\| \\ &\leq M \|f\| \quad \forall f \in X' \\ &\quad \forall x \in E \end{aligned}$$

Converse . Bidual .

$X \rightarrow$ nls .

$X' \rightarrow$ dual space .

$$X'' = (X')' \quad \text{bidual.}$$

$$F: X' \rightarrow \mathbb{K}$$

Reflexive space.

$$X \cong X''$$

$$1 < p, q < \infty$$

$$\frac{1}{p} + \frac{1}{q} = 1$$

$$(l^p)' \cong l^q$$

l^p is a reflexive

- l^1 is not reflexive. space
 $(l^1)' \cong l^\infty$
But $(l^\infty)' \not\cong l^1$

The. Let X be a normed linear space, then X is isometrically isomorphic to a

subspace of X'' .

Proof: For each (fixed)
 $x \in X$ define

$$F_x(f) = f(x) \quad f \in X'.$$

F_x is well defined &
linear.

$$|F_x(f)| = |f(x)| \leq \|f\| \|x\|$$

$\forall f \in X'$

- Then F_x is continuous

$$\& \quad \|F_x\| \leq \|x\|$$

(Con) Let $x \neq 0$. $\left[\|F_x\| = \sup \left\{ \frac{\|F_x(f)\|}{\|f\|} \right\} \right]$

& $x_0 \in X$ $x_0 \neq 0$. Then

$\exists f \neq 0$ s. t. $f \in X'$

& $\|f\| = 1$ & $f(x_0) = \|x_0\|$

$$\exists f \in X' \text{ s.t.}$$

$$f(x) = \|x\|, \quad \|f\| = 1$$

$$\therefore \quad \underline{\|F_x\| = \|x\|}$$

$$\text{For } x \in X$$

$$F_x \in X''$$

s.t

$$\|F_x\| = \|x\|$$

Define

$$\underline{T: X \rightarrow X''}$$

$$T(x) = F_x.$$

$$\therefore X \cong T(X) \subseteq X''$$

Seperability

If X' is seperable
then X is seperable.

$C \cap Y \subseteq X$
proper closed

$$x_0 \in X \setminus Y$$

$$\exists \tilde{f} \in X \text{ s.t.}$$

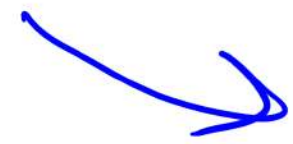
$$\tilde{f}|_Y = 0$$

$$\tilde{f}(x_0)$$

$$= \delta$$

$$= \text{dist}(x_0, Y)$$

$$\|\tilde{f}\| = 1$$

- subspace $\subseteq$ separable space

 separable.

X' is separable.

$$U = \{f \in X' \mid \|f\| = 1\}$$

$\Rightarrow U$ is separable.

Let $\{f_n\}$ be a
countable dense set.

$$\subseteq U, \quad \|f_n\| = 1 \quad \text{let } u_n.$$

Pick $\{\underline{x_n}\} \subseteq X$ s.t

$$\|x_n\|=1 \quad \& \quad |f_n(x_n)| \geq \frac{1}{2}$$

$$M = \overline{\text{span } \{x_n\}} \subseteq X$$

Suppose. $M \subsetneq X$
closed.

$$\exists g \in X' \text{ s.t. } \|g\|=1 \\ \& \quad g|_M = 0$$

$$\Rightarrow \textcircled{g \in \mathcal{U}} \text{ s.t. } g|_M = 0$$

Consider .

$$\frac{1}{2} \leq |f_n(x_n)|$$

$$= |f_n(x_n) - g(x_n)|$$

$$\leq \|f_n - g\| \|x_n\|$$

$$= \|f_n - g\| \quad \downarrow$$

$$\boxed{M = X}$$